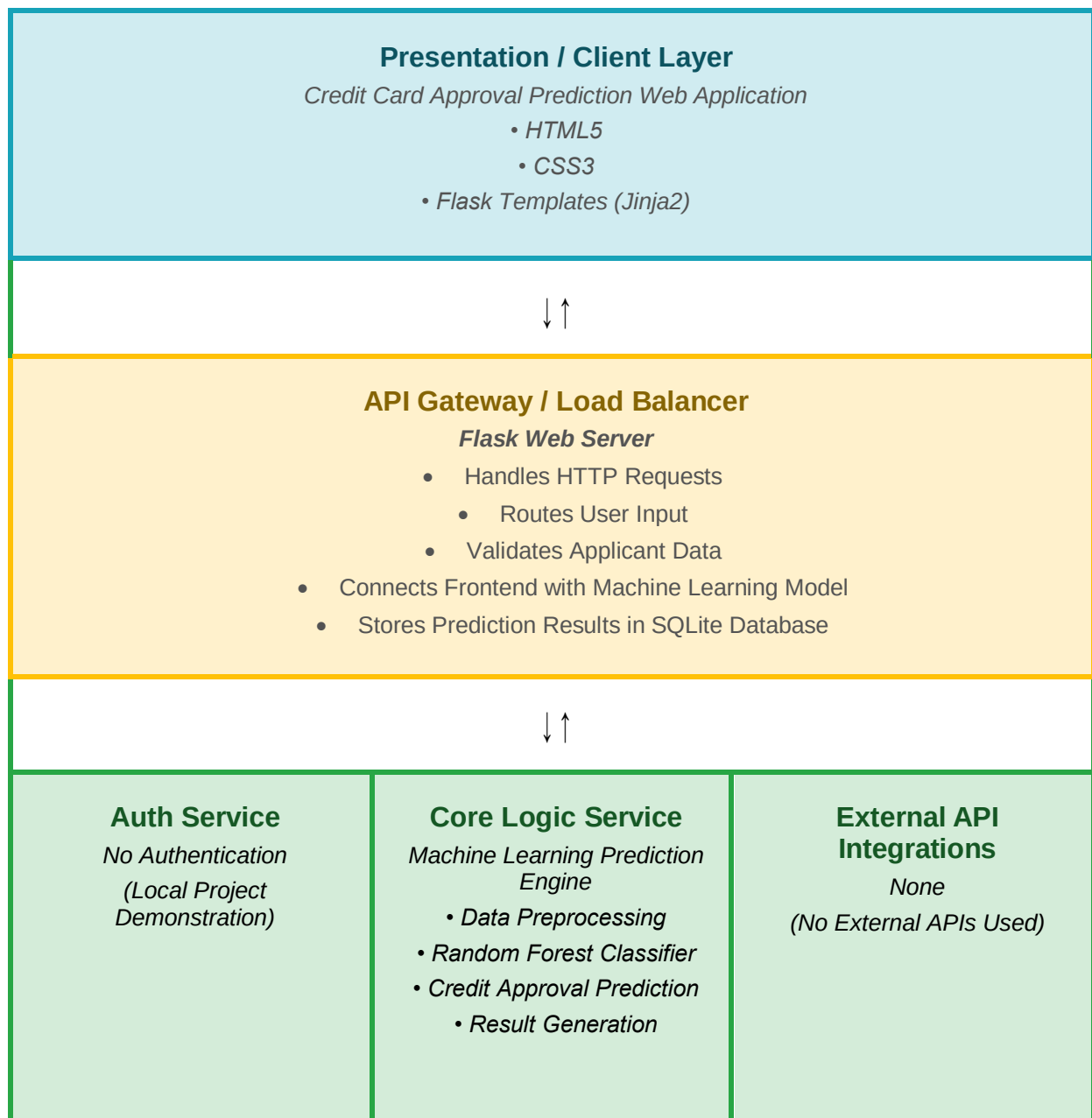


Solution Architecture

Date	5 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:





Data / Storage Layer

Dataset Storage

- *application_record.csv*
- *credit_record.csv*

Model Storage

- *model.pkl*

Database Storage

SQLite Database (***credit_card.db***)

Stores:

- *Applicant Name*
- *Prediction Result*
- *Confidence Score*
- *Prediction Date*
- *Prediction Time*
- *Prediction History*



Analytics Layer

Dashboard & Reporting

- *Total Predictions*
- *Approved Customers*
- *Rejected Customers*
- *Approval Rate*
- *Prediction History*

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface for entering applicant details, displaying prediction results, prediction history, and dashboard analytics.	HTML5, CSS3, Flask Templates
Application Layer	Receives user requests, validates inputs, loads the trained machine learning model, stores prediction results in SQLite, and returns prediction responses.	Python, Flask
Machine Learning Engine	Performs data preprocessing, feature encoding, prediction, confidence score calculation, and generates approval or rejection results using the Random Forest model.	Scikit-learn, NumPy, Pandas
Database	Stores applicant prediction history, confidence score, prediction date and time, and retrieves records for history and dashboard pages.	CSV Files, Pickle (.pkl)
Data & Model Storage	Stores training datasets and the serialized machine learning model used for prediction.	CSV Files, Pickle (.pkl)
Dashboard & Analytics	Displays prediction statistics such as total predictions, approvals, rejections, approval rate, and prediction history.	Flask, SQLite3